



pdf-tree — QA Sprint Backlog

Manual Validation & Tree Confidence Index Program

Field	Value
Document owner	DataClever AI — Engineering
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Assignee role	Junior AI / Software Support Engineer
Companion reference	docs/EVALUATION_GUIDE.md (read chapters 4–5 before starting Story 1)
Classification	Internal — Proprietary. Not for external distribution. See LICENSE.md.

Sprint Goal

Establish a repeatable, evidence-based QA process for pdf-tree output: a manually-validated, statistically-sampled inspection procedure for one document, generalized into a **Tree Confidence Index** computed across a 6-manual benchmark set. At sprint close, Engineering receives a scored confidence rating per manual and a consolidated, severity-classified bug list.

This backlog contains two stories. Story 2 depends on Story 1's method being executed correctly at least once — do not start Story 2 until Story 1 has been reviewed and approved by your lead.

Story 1 — Sampled Manual Structural Validation (Single Manual)

Objective: manually validate a statistically representative sample of a technical manual's `tree.json` output against the source PDF, following the procedure in `EVALUATION_GUIDE.md` §5.

Input: one structured technical manual PDF (e.g. a 280-page document, provided by your lead).

Task 1.1 — Environment Setup

Follow `EVALUATION_GUIDE.md` §5.1:

```
uv sync
uv run streamlit run streamlit_app/app.py
```

Capture the console output to a file for later reference — there is currently no persistent log file (§5.2 gap):

```
uv run streamlit run streamlit_app/app.py 2>&1 | tee run.log
```

Done when: the app is running locally and `run.log` is being written.

Task 1.2 — Run the Pipeline

On the **Extraction** page, upload the manual and run the full pipeline with Docling enabled. On completion:

- Record `total_pages` (from the **Metrics** page).
- Export `tree.json`, `images_v1.json`, and `bookmarks.json` from the **Export** page — these are your offline evidence set.
- Note the overall **Coverage / Precision / Structure** verdict from §5.3 (Metrics page). A FAIL here does not block sampling — it is itself input to your findings.

Done when: `tree.json` + companion exports are saved to your working folder, named `<manual_id>_tree.json` etc.

Task 1.3 — Compute the Sample

The sample size is **15% of total pages**, rounded up.

Example	Formula	Result
280-page manual	<code>ceil(280 × 0.15)</code>	42 pages

Sampling method — do not simply take the first 15% of pages. Use stratified sampling so early, middle, and late chapters are all represented, plus mandatory inclusion of the highest-risk sections regardless of quota:

1. From `tree.json`, list all top-level (chapter) sections and their page ranges.
2. Distribute the 42-page quota proportionally across chapters, by each chapter's share of `total_pages` (a 60-page chapter in a 280-page manual gets $\text{round}(60/280 \times 42) \approx 9$ sampled pages).
3. Within each chapter, pick pages at even intervals (e.g. every *N*th page) rather than clustering — this maximizes distinct-section coverage per chapter.
4. **Mandatory additions, on top of the 15% quota, not counted against it:**
 - Every section with `flagged_for_review: true`.
 - Every section with `structural_source: "inferred"`.
 - The first and last content section of the document (front-matter and back-matter boundary is the most common miscalibration point — see `EVALUATION_GUIDE.md` §4.2).

Document the final sampled page list and the section each page maps to (via `page_start` / `page_end`) before inspecting anything — this list is a deliverable, not scratch work.

Done when: a sample page/section list exists, reviewable independently of your findings (so a second engineer could re-run the same sample).

Task 1.4 — Manual Inspection

For **every section** touched by your sample (a page may map to only one section — inspect that section once, not once per page), run the full checklist from `EVALUATION_GUIDE.md`:

- §5.4 — Manual Structural Review Checklist (7 items: `hierarchy_level`, `parent_section_id/child_sections`, `page_start/page_end`, `hierarchy_path`, `gaps/duplicates`, `flagged_for_review`, `structural_source`).
- §5.5 — Embedded Image Extraction, for any sampled section containing images (3 items).
- §5.6 — Additional Field-Level Checks (`numbering_scheme`, `offset_applied`, table node content, `sanity_report`) (4 items).

For each item, open the source PDF at the relevant page(s) side-by-side and confirm against the checklist question. Record a **PASS** or **FAIL** — do not record “looks fine” without checking the specific page.

Done when: every sampled section has a checklist result recorded for every applicable item (not just a subset).

Task 1.5 — Log Findings

Use this column schema (spreadsheet or CSV) — one row per checklist item evaluated:

Column	Description
manual_id	Identifier for the source PDF
section_id	From tree.json
page_sampled	The page number that triggered inspection of this section
checklist_ref	e.g. 5.4–page_start_end, 5.6–numbering_scheme
result	PASS / FAIL
severity	See §Severity Classification below — required for every FAIL
evidence	Short description or screenshot reference (PDF page vs. tree.json field)
notes	Free text — root-cause hypothesis if apparent (e.g. “offset miscalibration, see toc_resolution.py”)

Done when: the log has one row per (section × checklist item) evaluated, with zero blank result cells.

Task 1.6 — Story 1 Deliverables

- Sample selection document (Task 1.3).
- Findings log (Task 1.5), all rows complete.
- Copy of the automated validation report (§5.3, Metrics page) for the manual.
- run.log console capture.
- Short summary (≤ 10 lines): total items checked, pass rate, count of FAIL by severity, and whether the automated PASS/FAIL verdict agreed with your manual sample.

Definition of Done: all five artifacts above exist, and your lead has reviewed the sample list (Task 1.3) and confirmed the methodology was followed before this story is marked complete.

Severity Classification (used in Story 1 and Story 2)

Apply this scale to every FAIL logged. This classification also drives the Confidence Index weighting in Story 2 — classify consistently.

Severity	Definition	Example
Critical	Content is misplaced into the wrong section, lost entirely, or the hierarchy is broken (cycle, missing parent, wrong nesting)	A subsection’s content assigned to its sibling; parent_section_id pointing at the wrong chapter
High	A flagged_for_review section is confirmed genuinely wrong on manual check, or page_start/page_end is off by more than 1 page	Section boundary two pages off due to offset miscalibration
Medium	Content is correctly placed but a supporting field is wrong or degraded	Image assigned to the wrong sibling section on a shared page; offset_applied inconsistent between adjacent sections

Severity	Definition	Example
Low	Cosmetic or non-blocking defect that does not affect retrievability of the correct content	Empty table canonical_text placeholder where the surrounding paragraph text still carries the information

Story 2 — Tree Confidence Index (6-Manual Benchmark)

Objective: repeat Story 1's method across **6 manuals** and produce a single, comparable **Confidence Index** score per manual, plus a consolidated bug report.

Precondition: Story 1 completed and approved for at least one manual, using the exact same checklist, severity scale, and sampling method for all 6.

Task 2.1 — Run Story 1's Full Procedure on Each of the 6 Manuals

Repeat Tasks 1.1–1.5 independently for each manual. Each manual gets its own sample size ($\text{ceil}(\text{total_pages} \times 0.15)$), its own sample list, and its own findings log — do not reuse or pool samples across manuals.

Done when: 6 complete findings logs exist, one per manual, each satisfying Story 1's Definition of Done.

Task 2.2 — Compute the Confidence Index per Manual

Formula:

Confidence Index = $100 \times (1 - (\text{sum}(\text{severity_weight} \times \text{fail_count}) / \text{total_items_evaluated}))$
clamped to the range [0, 100].

Severity weights:

Severity	Weight
Critical	8
High	4
Medium	2
Low	1

Automated-check override: if the automated validation report (§5.3) for a manual shows status: FAIL on **Coverage** or **Structure** (not Precision alone), cap that manual's Confidence Index at **60**, regardless of the formula result. Rationale: these two checks indicate content loss or a broken hierarchy — a systemic defect the manual sample size may not fully capture, and the score should reflect that risk rather than understate it.

Worked example: 42-page sample maps to 30 distinct sections, each checked against 11 checklist items -> $\text{total_items_evaluated} = 330$. Findings: 1 Critical, 2 High, 4 Medium, 3 Low.

$\text{weighted_fails} = (1 \times 8) + (2 \times 4) + (4 \times 2) + (3 \times 1) = 8 + 8 + 8 + 3 = 27$
Confidence Index = $100 \times (1 - 27/330) = 100 \times 0.918 = 91.8 \rightarrow 92$

Done when: each of the 6 manuals has one computed Confidence Index score, with the calculation shown (not just the final number).

Task 2.3 — Consolidate the Bug List

Merge all FAIL rows across the 6 findings logs into one list. Deduplicate by root cause, not by row — three sections all misplaced by the same offset-calibration bug are **one bug entry** referencing three affected sections, not three separate bugs.

Column schema for the consolidated list:

Column	Description
bug_id	Sequential identifier
title	One-line description
severity	Highest severity among affected instances
manuals_affected	Which of the 6 manuals show this bug
sections_affected	Section IDs / count
suspected_module	Pipeline module, if identifiable (e.g. <code>toc_resolution.py</code> , <code>section_matcher.py</code> — see <code>EVALUATION_GUIDE.md §4.1</code>)
escalate	Yes/No per <code>EVALUATION_GUIDE.md §7 Escalation Guidance</code>

Done when: every FAIL row from all 6 logs is accounted for in exactly one consolidated bug entry.

Task 2.4 — Sprint Deliverable: Confidence Index Report

One summary table, plus the consolidated bug list:

Manual	Total Pages	Sample Size	Automated Verdict	Confidence Index	Critical	High	Medium	Low
Manual 1	—	—	—	—	—	—	—	—
Manual 2	—	—	—	—	—	—	—	—
Manual 3	—	—	—	—	—	—	—	—
Manual 4	—	—	—	—	—	—	—	—
Manual 5	—	—	—	—	—	—	—	—
Manual 6	—	—	—	—	—	—	—	—

Plus: - The consolidated, deduplicated bug list (Task 2.3). - A one-paragraph interpretation: is the Confidence Index stable across manuals, or does it vary sharply by document type (e.g. lower on manuals with `structural_source: "inferred"`)? This observation is often more actionable to Engineering than the raw scores.

Definition of Done (Story 2): the summary table is fully populated, the consolidated bug list references every finding, and the report has been reviewed with your lead before being shared with Engineering.

Overall Sprint Definition of Done

- ☐ Story 1 completed, reviewed, and approved for the pilot manual.
- ☐ Story 2 completed for all 6 manuals using the identical method from Story 1.
- ☐ Confidence Index Report delivered (Task 2.4).
- ☐ Every Critical and High severity bug has an escalation decision recorded (`EVALUATION_GUIDE.md §7`).
- ☐ All raw artifacts (sample lists, findings logs, `tree.json` exports, `run.log` captures) retained and linked from the final report — the Confidence Index number must be traceable back to the individual checklist rows that produced it.

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